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Graduation project

SYSTEM ARCHITECTURE

PLANNING STAGE

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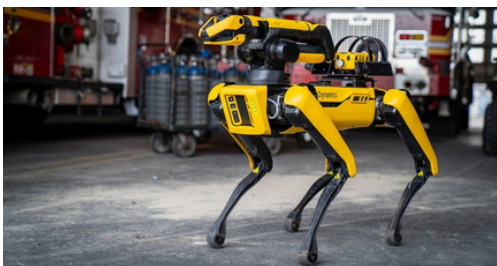
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Introduction:

A mobile robot is a machine controlled by software that use sensors and other technology to identify its surroundings and move around its environment. Mobile robots function using a combination of artificial intelligence (AI) and Mechanical robotic elements, such as wheels, tracks and legs. Mobile robots are becoming increasingly popular across different business sectors. They are used to assist with work processes and even accomplish tasks that are impossible or dangerous for human workers.

Mobile robots can be classified in two ways: by the environment in which they work and by the device they use to move. There are also two main types of mobile robots: autonomous and non-autonomous, or guided, mobile robots. Guided mobile robots require some form of instruction or guidance system in order to move, whereas autonomous mobile robots (AMRs) are able to move and explore their surroundings without any sort of external direction.



Mobile legged robot from boston Dynamics



Mobile tracked robot



Mobile wheeled robot for construction sites

Literature Review:

Autonomous Guided Vehicles (AGVs) are earlier forms of mobile robots used in industrial settings for material handling and transportation of goods in warehouses. They navigate using predefined paths marked by magnetic strips, wires, or optical lines embedded in the floor. This fixed-path navigation limits their flexibility. AGVs use wire-guided, magnetic tape, optical guidance, or laser-guided systems to follow these paths. While AGVs can operate without direct human intervention, any changes in the path or environment require manual reconfiguration. They are not well-suited to dynamic environments, as unexpected obstacles can disrupt their operation.

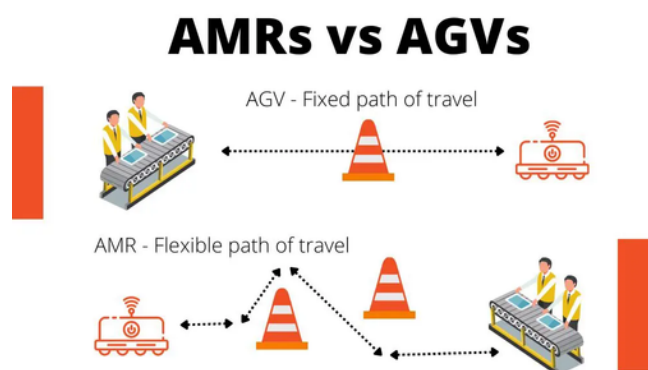
In contrast, Autonomous Mobile Robots (AMRs) use advanced navigation systems, including LIDAR, cameras, and ultrasonic sensors, to understand and navigate their environment dynamically. They use simultaneous localization and mapping (SLAM) to build maps of unknown environments while keeping track of their location. AMRs are highly autonomous, making independent decisions based on sensor inputs and onboard algorithms, allowing them to navigate complex and changing environments without predefined paths.

AMRs are equipped with artificial intelligence and machine learning capabilities, enabling them to learn from their environment and improve their navigation and task execution over time. They are highly flexible and can adapt to changes in their environment, rerouting in real-time to avoid obstacles and find the most efficient path. This adaptability makes AMRs suitable for a wide range of tasks in various environments, including warehouses, hospitals, and public spaces.

The main advantages of AMRs over AGVs are reduced infrastructure costs, improved efficiency, and enhanced safety. AMRs do not require extensive physical infrastructure, making them more cost-effective to implement and reconfigure. They can optimize their paths and tasks dynamically, leading to better efficiency and productivity. Their advanced sensors and navigation algorithms allow them to avoid obstacles and reduce the risk of accidents, enhancing safety in busy environments. While AGVs provide reliable solutions for controlled environments with predefined tasks, AMRs offer superior flexibility and autonomy for dynamic and complex settings.



AGV that follows a line for material handling



AMR VS AGV in Navigation



AMR Navigation without lines or strips



AGV that follows a magnetic strip

Problem Statement:

Managers across various industries, including life sciences, healthcare, manufacturing, and warehousing, often face the challenge of choosing between increasing labor costs for repetitive tasks or diverting skilled staff from their primary responsibilities. High turnover rates among low-skill, part-time workers lead to ongoing labor and training expenses, while using experienced staff for menial tasks lowers productivity, morale, and focus on high-value work, ultimately increasing human effort and causing burnout.

In warehousing, the manual process of cross-docking—directly transferring goods from receiving to shipping docks—requires precise coordination to avoid delays, misrouting, and increased labor costs. Inefficiencies in this process result in longer lead times, potential supply chain bottlenecks, incorrect shipments, customer dissatisfaction, and higher operational costs due to the need for additional labor.

Objectives:

This project aims to address and solve these challenges by developing an Autonomous Mobile Robot (AMR) designed specifically for the task of transferring goods within a warehouse. The AMR will be equipped with advanced navigation and control systems to autonomously move through the warehouse, handle varying load sizes, and integrate seamlessly with existing warehouse management systems. By implementing this AMR, the project seeks to improve operational efficiency, reduce labor costs, and enhance safety within warehouse environments.

Phases:

- **Phase 1 (Research and Planning):** establish a comprehensive understanding of the requirements, current technologies, and methodologies relevant to the development of an Autonomous Mobile Robot (AMR) for transferring goods within a warehouse. Analyze existing solutions and identify their strengths and limitations.
- Define the functional and performance requirements for the AMR.
- Identify and document the specific tasks the AMR will perform within the warehouse.
- Research the latest advancements in sensors, actuators, control systems, and navigation algorithms.
- Determine the necessary hardware components, such as sensors, motors, controllers, and power systems.
- Select appropriate software tools and development environments, including programming languages, simulation software, and version control systems.
- Establish an estimated budget for the project, covering hardware, software, labor, and any other associated costs.
- Develop initial design concepts for the AMR's mechanical structure, considering factors such as load capacity, stability, and mobility.
- Outline the electrical systems, including power distribution, sensor integration, and communication networks.
- Design the software architecture, focusing on navigation, control algorithms, and user interfaces.
- Build a simple prototype, such as an RC car, to serve as a test platform for implementing and testing software algorithms and methodologies.
- Allocate approximately **two and a half months** to complete this phase, ensuring a thorough and detailed preparation for the subsequent phases.

Phase 2 (Design and Development): Transform the initial concepts and requirements gathered in Phase 1 into detailed designs and begin the development of the mechanical, electrical, and software components of the Autonomous Mobile Robot (AMR). This phase should include the following:

- Develop the design of the robot chassis, ensuring it meets structural requirements for durability and stability.
- Design the load handling mechanism, including any lifting or carrying devices.
- Perform stress and load analysis to ensure the chassis can withstand operational conditions.
- Design the mobility system, including the selection of wheels, tracks, or other mobility mechanisms.
- Calculate power requirements and select appropriate motors and drive systems.
- Incorporate safety features such as emergency stop buttons and fail-safes.
- Ensure the design maintains stability during operation, especially when handling loads.
- Create the power distribution system, including batteries and power regulation circuits.
- Ensure the power system meets operational requirements for runtime and load handling.
- Develop detailed circuit diagrams for all electronic components, including motor controllers, sensors, and communication modules.
- program and Simulate circuits to test their functionality.
- Select and integrate sensors required for navigation, obstacle detection, and load handling.
- Ensure proper calibration and positioning of sensors for optimal performance.
- Develop algorithms for real-time navigation and mapping, allowing the AMR to autonomously navigate the warehouse environment.

- Implement simultaneous localization and mapping (SLAM) techniques to build and update maps.
- Simulate the ROS algorithm on a small RC car.
- Assemble the mechanical and electrical components to build the initial prototype of the AMR.
- Ensure all components are securely fitted and connected.
- Conduct initial tests to verify the functionality of the mechanical, electrical, and software components.
- Identify and troubleshoot any issues that arise during the initial testing phase.
- Timeline: Allocate approximately **three months** to complete this phase, ensuring detailed designs are developed and initial prototypes are built and tested effectively.

Phase 3 (Prototype Development): Build and refine a functional prototype of the Autonomous Mobile Robot (AMR), integrating all mechanical, electrical, and software components developed in the previous phases. This phase aims to validate the design through iterative testing and refinement, ensuring the prototype meets the project's performance and reliability requirements.

- Fabricate and assemble the chassis, load handling mechanism, and mobility system components.
- Ensure all mechanical parts fit together accurately and securely.
- Use precise manufacturing techniques to meet the design specifications.
- Install and wire all electronic components, including sensors, motor controllers, and communication modules.
- Ensure proper power distribution and regulation through the designed power management system.
- Calibrate sensors and verify their integration for optimal performance.

- Integrate navigation, mapping, obstacle detection, path planning, and control algorithms.
- Ensure the Robot Operating System (ROS) nodes are properly set up and communicating with each other.
- Conduct initial power-on tests to verify the basic functionality of the assembled prototype.
- Check for any issues with power distribution, component connections, and software initialization.
- Address any initial problems to ensure the prototype is ready for more extensive testing.
- Evaluate the structural integrity, load handling capability, and stability of the AMR.
- Make any necessary adjustments to ensure mechanical reliability.
- Identify any issues or areas for improvement during testing.
- Make iterative adjustments to the mechanical, electrical, and software components to address these issues.
- Re-test the prototype after each adjustment to ensure improvements are effective.
- Document the assembly process, including detailed steps, tools used, and any challenges encountered.
- Record the results of all tests and validation activities.
- Update design documents, circuit diagrams, and software documentation to reflect any changes made during prototype development.
- Timeline: Allocate approximately **four months** to complete this phase

Phase 4 (Final Testing and Evaluation): Thoroughly evaluate the fully assembled prototype of the Autonomous Mobile Robot (AMR) under real-world conditions to ensure it meets all performance, reliability, and safety requirements. This phase aims to identify any remaining issues, validate the system's functionality, and assess its readiness for deployment.

- Conduct a series of comprehensive tests to evaluate the AMR's overall performance.
- Test all integrated systems, including navigation, obstacle detection, load handling, and power management, in various operational scenarios.
- Measure key performance metrics such as speed, accuracy, battery life, load capacity, and efficiency.
- Compare these metrics against the project's initial objectives and requirements to ensure the AMR meets or exceeds expectations.
- Conduct a final round of validation tests to confirm that all issues have been resolved and the AMR is ready for deployment.
- Ensure the system meets all performance, reliability, and safety criteria.
- Document all testing procedures, results, and any changes made during the final testing phase.
- Update design documents and test reports.
- Timeline: Allocate approximately **Two months** to complete this phase

Overview of Autonomous Mobile Robots

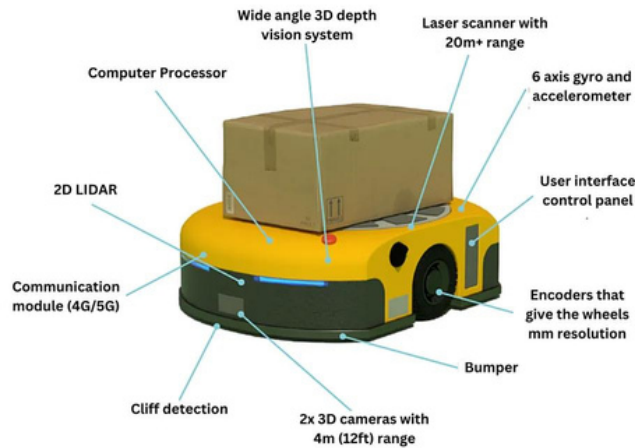
Technologies in AMRs:

Navigation and Mapping: Simultaneous Localization and Mapping (SLAM) is a fundamental technology that allows AMRs to create a map of an unknown environment while simultaneously keeping track of their location within that environment. SLAM algorithms use data from various sensors to build and update maps in real-time. The Global Positioning System (GPS) primarily used for outdoor navigation can assist AMRs in determining their precise location in open environments, though it is less effective indoors due to signal interference. Inertial Measurement Units (IMUs) provide data on the robot's orientation, acceleration, and angular velocity, crucial for maintaining stability and aiding navigation algorithms.

Sensors: LiDAR (Light Detection and Ranging) sensors emit laser pulses to measure distances to surrounding objects, creating detailed 3D maps of the environment. LiDAR is essential for accurate navigation and obstacle detection. Vision systems using monocular, stereo, or depth cameras capture visual information from the environment, used for object recognition, navigation, and interaction with human operators. Ultrasonic and Infrared Sensors detect nearby objects by measuring the time it takes for emitted sound waves or infrared light to bounce back, often used for close-range obstacle detection and avoidance.

Control Systems: Microcontrollers and Embedded Systems control the robot's hardware components, such as motors and sensors, executing control algorithms to ensure real-time responsiveness. The Robot Operating System (ROS) is an open-source framework for writing robot software, providing tools and libraries for developing and managing robot applications, facilitating communication between different components of the AMR.

Human-Robot Interaction: A Graphical User Interface (GUI) is a visual interface that allows users to interact with a system through graphical elements such as buttons, icons, and menus. It provides an intuitive and user-friendly way to control and monitor devices, including AMRs, without requiring knowledge of complex commands or programming.



AMR composition

By integrating these technologies, AMRs achieve high levels of autonomy, efficiency, and reliability, making them invaluable assets in modern warehousing and logistics operations.

Applications of AMR in warehouses: AMRs are used to transport goods within the warehouse, moving items from storage to packing areas, loading docks, or other locations. They efficiently handle repetitive tasks, such as moving pallets, bins, and containers, reducing the need for manual labor.

They're also equipped with advanced sensors and scanning technologies can perform inventory checks by moving through aisles and scanning barcodes or RFID tags. This helps maintain accurate inventory records, reduces errors, and allows for real-time inventory tracking.

AMRs can work alongside human workers, enhancing their productivity and safety. For example, they can deliver tools, parts, or supplies to workers on the warehouse floor, allowing humans to focus on more complex tasks.

Existing Solutions and Their Limitations: AGVs are pre-programmed to follow fixed paths using magnetic strips, wires, or tracks embedded in the warehouse floor.

Advantages:

- Automated transport along fixed paths.
- Reduces the need for manual labor.

Disadvantages:

- Lack of Flexibility: Restricted to fixed paths and cannot easily adapt to changes in the warehouse layout or dynamic obstacles.
- Complex Installation: Costly and time-consuming to install and maintain the guidance infrastructure.
- Limited Scalability: Requires significant modifications to the infrastructure for expansion.

Conveyor belts and roller conveyors are used to transport goods along fixed paths within the warehouse.

Advantages:

- Efficient transport of goods along fixed paths.
- Can handle high volumes of goods.

Disadvantages:

- Space Constraints: Requires significant floor space, limiting warehouse layout flexibility.
- High Initial Cost: Substantial upfront investment for installation.
- Lack of Flexibility: Not adaptable to changes in workflow or warehouse configuration and limited to linear paths.



Warehouse Conveyor system

Human workers perform picking, packing, and material handling tasks manually.

Advantages:

- High flexibility and adaptability to various tasks.
- No need for extensive infrastructure.

Disadvantages:

- Labor Costs: High expenses for wages, benefits, and training.
- Human Error: Prone to errors, affecting order accuracy and inventory management.
- Physical Strain: Repetitive tasks can lead to worker fatigue and injuries, impacting productivity and safety.

Robotic arms and fixed automation systems are used for tasks such as picking and sorting within defined areas.

Advantages:

- High precision in tasks like picking and sorting.
- Can operate continuously without fatigue.

Disadvantages:

- Limited Mobility: Restricted to specific areas, not able to move around the warehouse.
- High Cost: Prohibitive costs for installation and integration.
- Complex Programming: Requires specialized programming and maintenance, adding to operational complexity.

Manual and automated forklifts and pallet jacks are used for moving heavy pallets and goods.

Advantages:

- Effective for moving heavy pallets and goods.
- Versatile in various warehouse tasks.

- Limitations:
 - Operator Dependence: Manual forklifts require skilled operators, leading to increased labor costs and potential safety risks.
 - Limited Automation: Automated forklifts are less flexible than AMRs and often require predefined paths or operator assistance.



Fixed robotic arm



Forklifts and pallet jacks

Why AMRs are the Optimal Solution

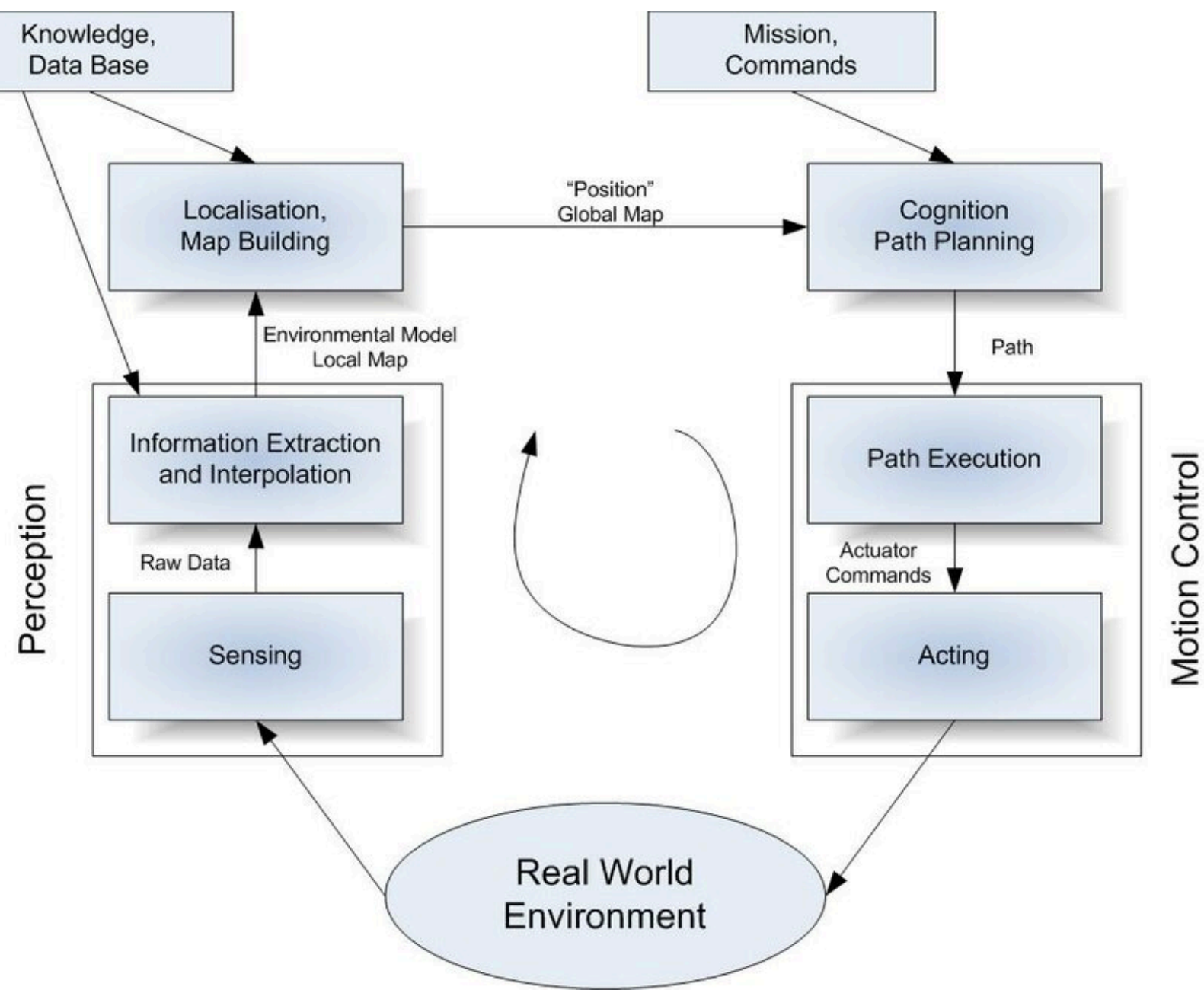
- Flexibility and Adaptability: AMRs navigate dynamically within the warehouse without needing fixed infrastructure, using advanced sensors and algorithms to adapt to changing layouts and obstacles. Suitable for various tasks and configurations.
- Scalability: Easily scalable by adding more units without significant infrastructure modifications. Allows gradual expansion and adaptation to increasing operational demands.
- Cost-Effectiveness: Despite substantial initial investment, AMRs offer long-term savings through reduced labor costs, increased efficiency, and lower maintenance expenses compared to fixed infrastructure solutions.
- Improved Accuracy and Efficiency: Utilize advanced navigation and picking technologies to enhance order accuracy and reduce errors, working continuously without fatigue for higher productivity and consistent performance.

- **Safety:** Equipped with sophisticated safety features like collision detection and avoidance, making them safer to operate around human workers and reducing the risk of workplace accidents and injuries.
- **Ease of Implementation:** Require minimal infrastructure changes, making them easier and quicker to implement compared to AGVs and conveyor systems, with minimal disruption to ongoing operations.
- **Versatility:** Programmable for various tasks, including goods transport, order picking, inventory management, and quality inspection, maximizing their utility within the warehouse.
- **Data Collection and Analytics:** Collect valuable operational data, which can be analyzed to optimize warehouse processes, improve decision-making, and enhance overall operational efficiency.

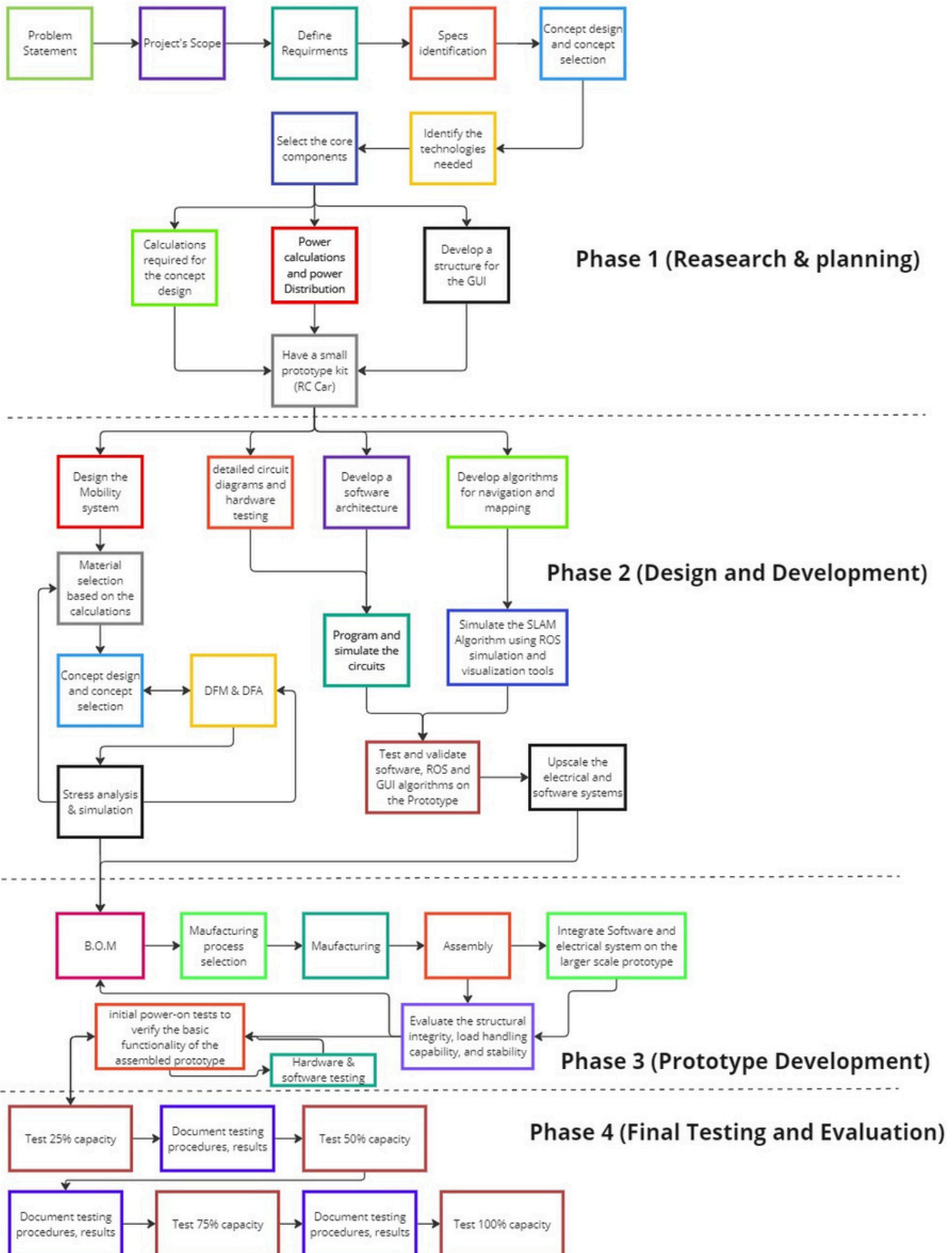
In conclusion, while existing solutions like AGVs, conveyors, and manual labor have their uses, they come with significant limitations in terms of flexibility, scalability, and cost. AMRs overcome these limitations, offering a more versatile, efficient, and cost-effective solution for modern warehouse operations.

System Design and Architecture

General Scheme of AMRs

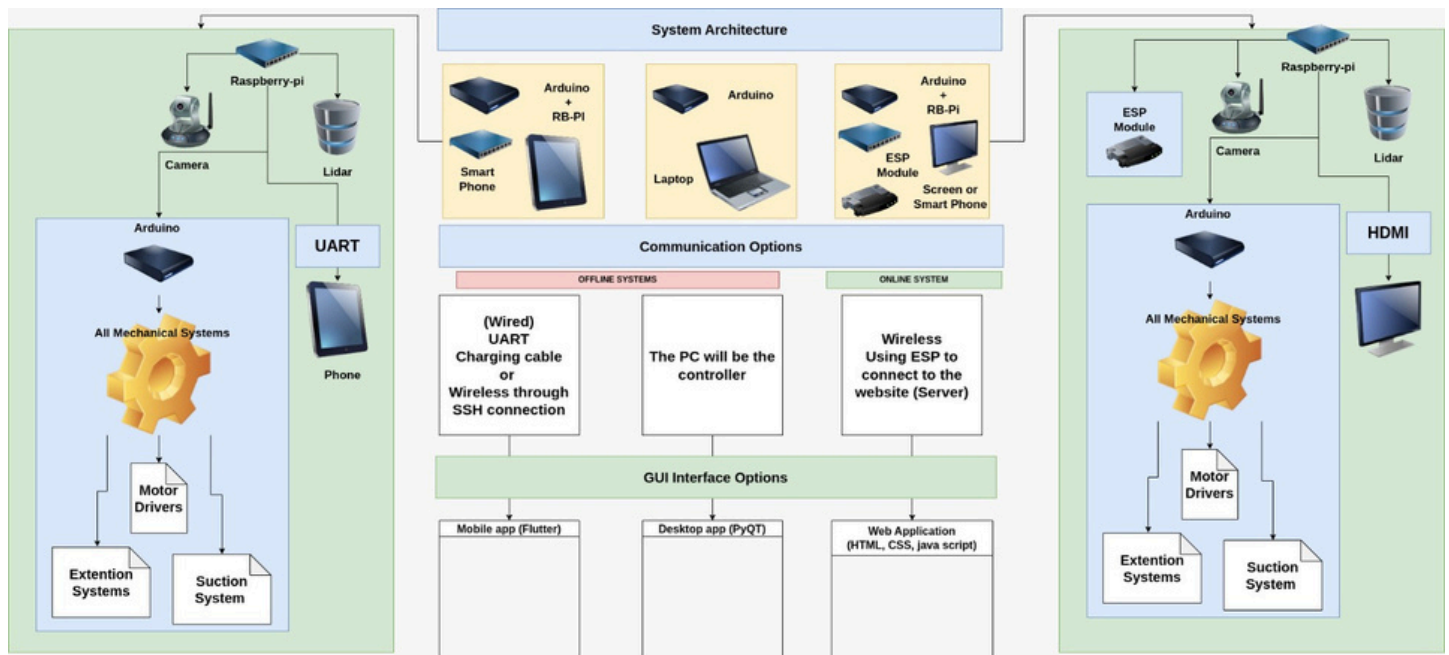


Overall System Architecture



GUI:

GUI Diagram



Benefits of a GUI for a robot

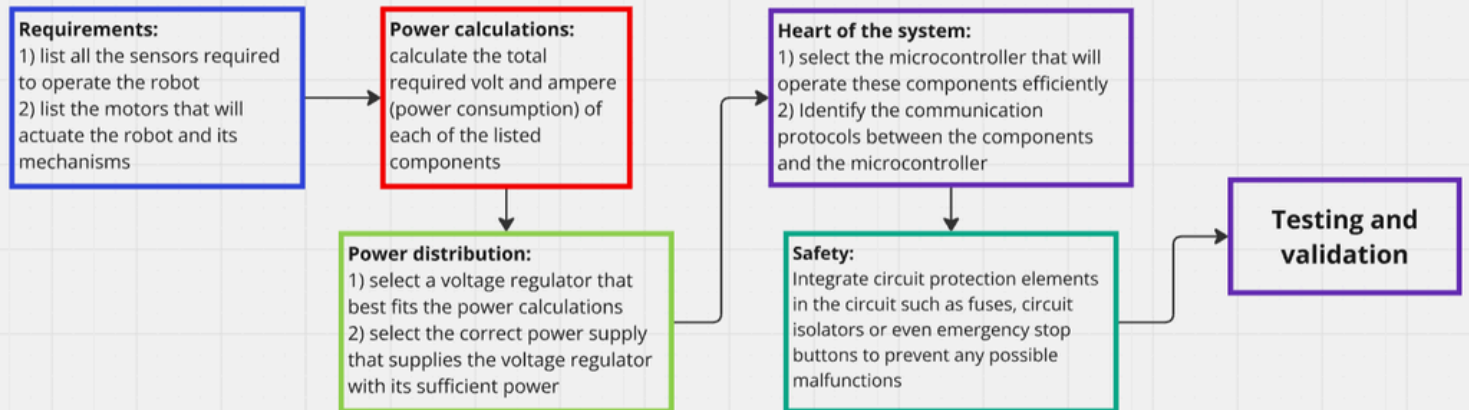
- **Ease of Use:** Users can interact with the robot through visual elements such as buttons, sliders, and menus, making it easier for those who may not be familiar with programming or technical commands.
- **Error Handling:** GUIs can display error messages and troubleshooting tips in a clear and accessible manner. This helps users to quickly identify and resolve issues without needing to sift through logs or error codes.
- **Enhanced Interaction:** For collaborative robots or robots that interact with humans, a GUI can provide a more engaging and interactive experience. Users can easily give commands and receive feedback, improving the overall interaction.

Possible GUI tools:

- Pyqt
- MIT GUI tool
- Pytinkter
- Qt-Creator

Electrical System:

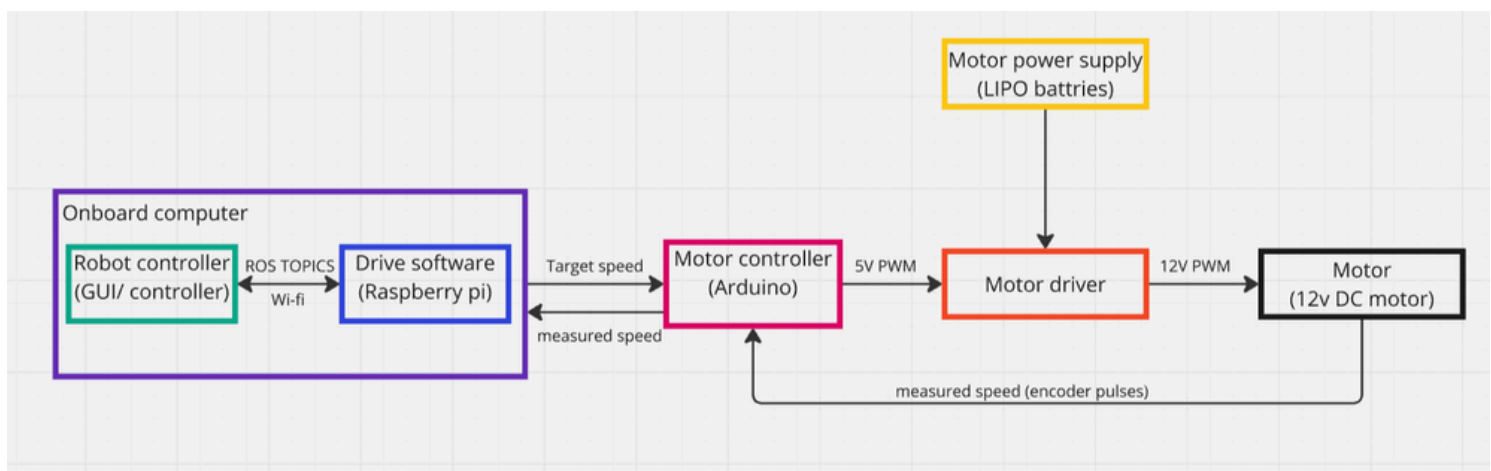
Electrical system flowchart



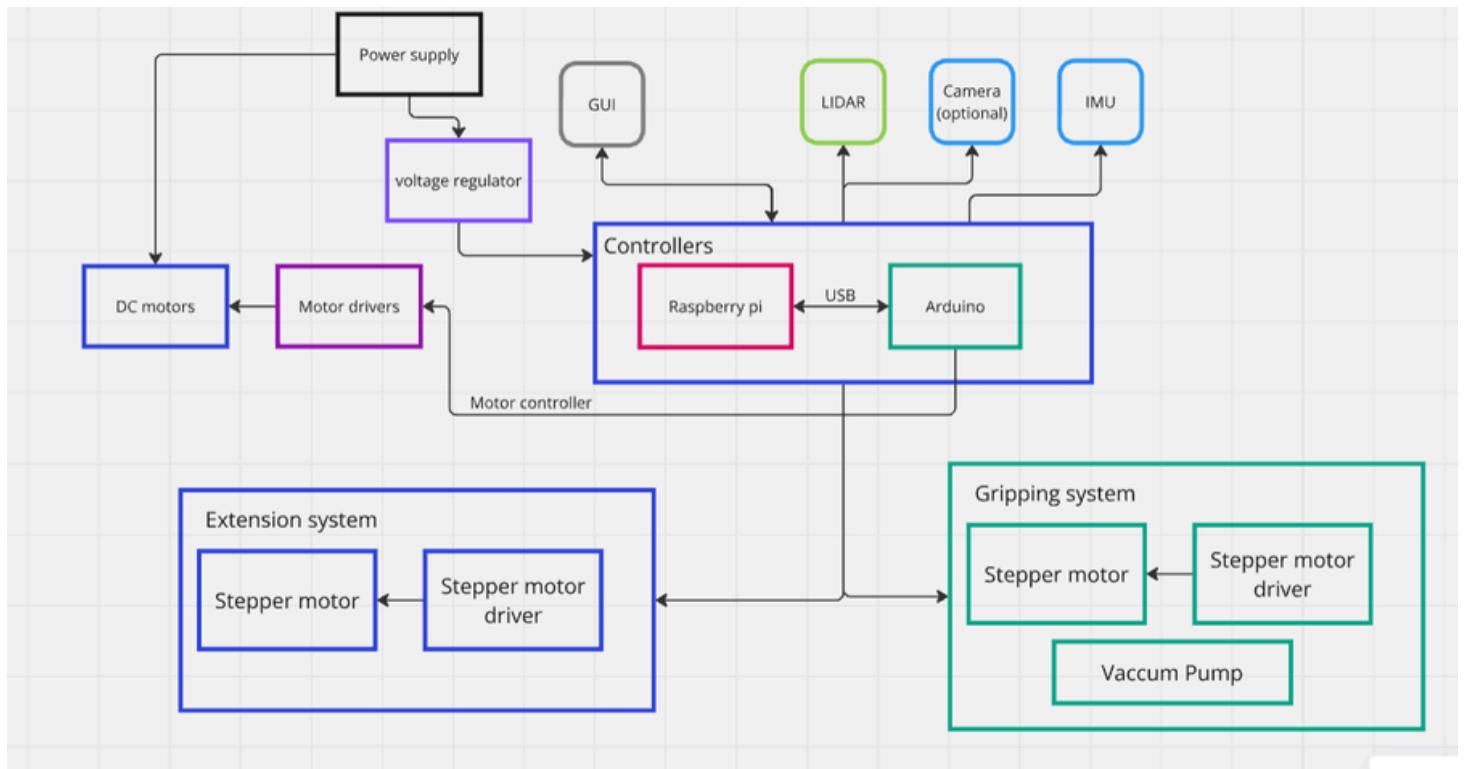
Core Components for the prototype:

- Lidar
- IMU
- Raspberry pi / Mini PC
- arduino
- DC motors

Motor diagram:

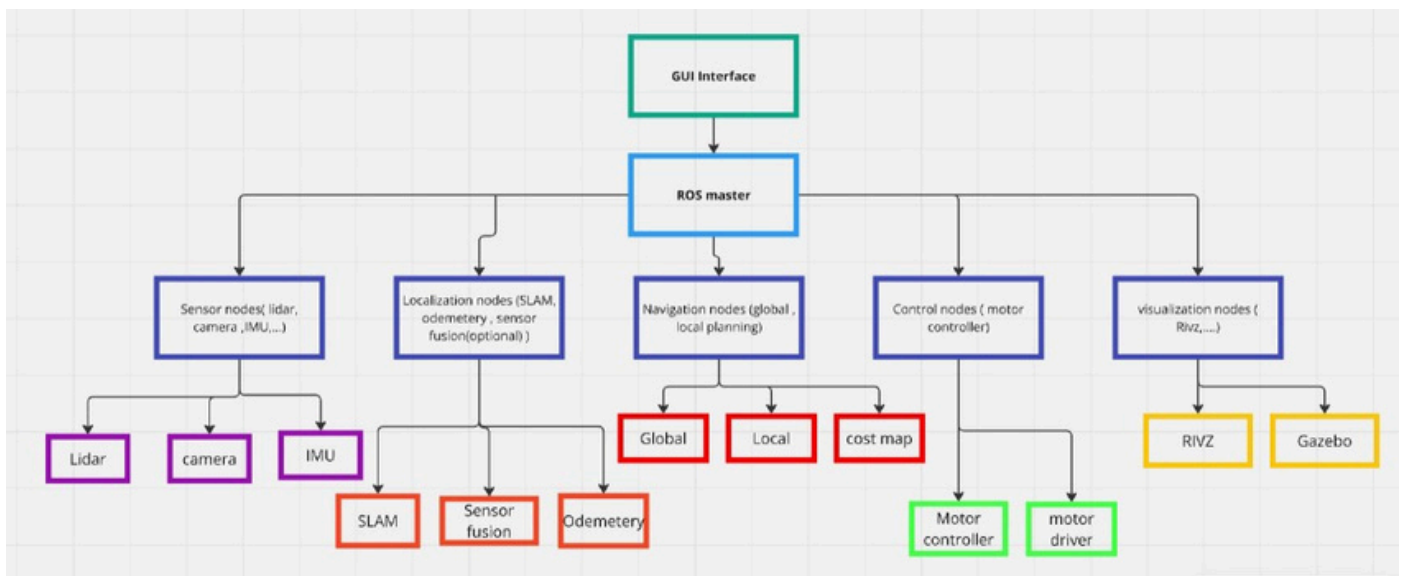


Electrical system of the AMR:



ROS:

ROS Diagram



Levels of Automation for a Mobile Robot:

Level 0: No Automation

- Description: The human operator performs all tasks.
- Example: A manually controlled forklift where the operator is responsible for all driving functions.

Level 1: Assisted Automation

- Description: The system assists the human operator with specific tasks but does not perform the full range of operations autonomously.
- Example: Basic automated guided vehicles (AGVs) that follow predefined paths and require human intervention for complex navigation or decision-making.

Level 2: Partial Automation

- Description: The system can perform some tasks autonomously but requires the human operator to be engaged and supervise the system at all times.
- Example: An autonomous mobile robot (AMR) that can navigate and avoid obstacles within a limited environment but needs human intervention for route planning or error correction.

Level 3: Conditional Automation

- Description: The system can perform most tasks autonomously in certain environments or under specific conditions, but human intervention is needed in complex scenarios.
- Example: An AMR in a structured warehouse environment that can navigate, avoid obstacles, and pick items autonomously but requires a human operator for complex decision-making or unstructured environments.

Level 4: High Automation

- Description: The system operates fully autonomously in specific environments or conditions, with no need for human intervention during normal operations. However, it may require human oversight or intervention in exceptional circumstances.
- Example: An AMR that operates autonomously in a controlled factory setting, performing tasks such as material handling or delivery without human assistance but may require help during maintenance or unexpected situations.

Level 5: Full Automation

- Description: The system is capable of performing all tasks autonomously in any environment and under any conditions that a human operator could manage.
- Example: A fully autonomous mobile robot that can navigate complex and unstructured environments, perform diverse tasks, and handle all decision-making processes independently without any human intervention.

How to Implement Automation

1. Mapping

- Description: Creating a map of the environment and localizing the robot within it simultaneously.
- SLAM (Simultaneous Localization and Mapping): Algorithms that simultaneously build a map of the environment and localize the robot within it.
 - Popular SLAM packages:
 - Gmapping
 - SLAM Toolbox

2. Localization

- Description: Determining the robot's position within a known map.
- Common Methods:
 - AMCL (Adaptive Monte Carlo Localization): A probabilistic localization system that uses a particle filter to track the robot's pose against a known map.
 - EKF (Extended Kalman Filter): Often used for sensor fusion, integrating data from wheel odometry, IMU, and GPS to estimate the robot's pose.

3. Path Planning

- Description: Determining a route from the robot's current location to the desired destination.
- Types of Path Planning:
 - Global Planning: Computes an overall path from start to goal, avoiding known obstacles.
 - Local Planning: Dynamically adjusts the path in response to moving obstacles using algorithms like the Dynamic Window Approach (DWA) or Trajectory Rollout.

4. Object Avoidance

- Description: Ensuring that the robot can navigate around obstacles in its environment.
- Common Strategies:
 - Costmaps: Represent obstacles and free space in the environment. The global costmap is used for global planning, and the local costmap is used for obstacle avoidance.
 - Inflation Layer: Adds a buffer around obstacles to ensure the robot maintains a safe distance.

Mapping Packages:

1. Hector SLAM

- Description: Hector SLAM fuses 2D SLAM and 3D navigation based on LiDAR scan data. It utilizes the high update rate of LiDAR sensors for real-time movement estimation.
- Strengths:
 - High Update Rate Dependency: Achieves real-time movement estimation using rapid LiDAR data.
 - Odometry Independence: Suitable for environments with unreliable odometry or dynamic elements.
- Weaknesses:
 - Low Scan Rates: Performance issues arise when LiDAR scan rates are low, as it cannot utilize accurate odometry information from wheeled robots.

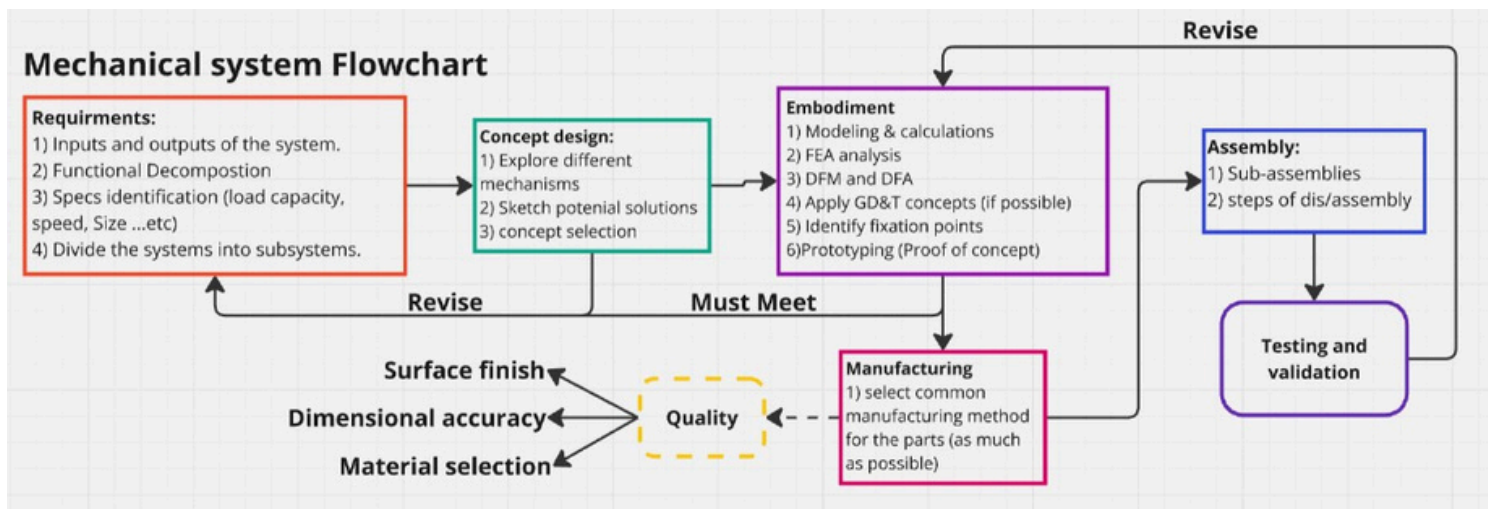
2. Gmapping

- Description: Gmapping is a ROS package providing a laser-based SLAM solution. It uses a Rao-Blackwellized Particle Filter (RBPF) to create 2D occupancy grid maps from laser scan data and odometry information.
- Strengths:
 - Odometry Integration: Increases mapping accuracy, decreases estimation error, and requires fewer particles, thus reducing computational effort.
- Weaknesses:
 - Wheel Slip Sensitivity: Performance may degrade significantly with wheel slip.
 - Dynamic Environments: Less effective in highly dynamic environments with many moving objects.

3. SLAM Toolbox

- **Description:** SLAM Toolbox uses both local and global approaches. Locally, it performs consistent pose estimation by matching new scans with recent scans. Globally, it matches the most recent scan against the map for loop closure.
- **Strengths:**
 - **Versatile Data Use:** Utilizes both laser scan data and odometry data.
 - **Multi-Session Mapping:** Supports multiple mapping sessions, building accurate maps in real-time.
 - **Flexible Mapping Modes:** Provides multiple modes of mapping and data serialization for enhanced functionality.

Mechanical System:



Design Requirements

Load capacity:

- Maximum weight: 60 KG
 - Package dimensions: Up to 40x60x30 (WxLxH) (cm) Slightly larger than the maximum package size
- Note (Height without the lifter)

Lifting Mechanism:

- Scissor lift: Must be able to lift up to 60 kg.
- Must provide stability during lifting and transport
- Adjustable height to accommodate various storage shelf heights, Max of 2m
- Linear actuators
- Material: High-strength steel

Picking Mechanism:

- Suction mechanism: Must securely hold packages up to 60 kg.
- Adjustable suction strength to handle different package surfaces.
- Multiple suction cups to distribute the load evenly
- Adjustable arm to position suction
- Material: Rubber or silicone for the suction cups.
- Actuation: Vacuum pump system with an adjustable control

Mobility:

- Autonomous navigation: Must be able to move autonomously within the warehouse, Omni-directional wheels
- Obstacle avoidance: Must detect and avoid obstacles,
- Speed: Must be efficient in transport but safe within the operational environment

Power source:

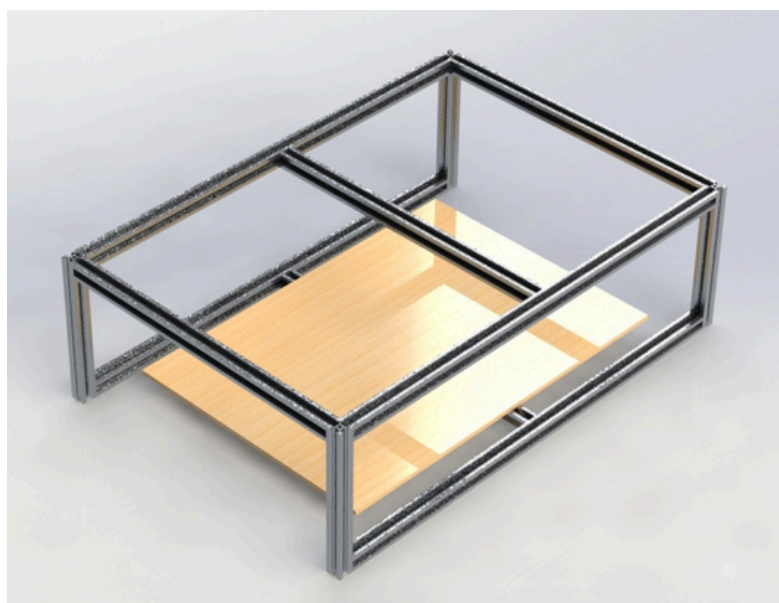
- Battery life: Sufficient for a full shift for example.
- Rechargeable: Quick and efficient recharging process.

Safety features:

- Emergency stop: Accessible and responsive emergency stop mechanisms.
- Weight sensors: To prevent overloading.
- Collision sensors: To avoid damaging goods or causing accidents.
- Weight Sensors: Placed on the lifting platform to monitor the load.
- Collision Sensors: LIDAR or camera sensors for obstacle detection.
- Emergency Stop: Easily accessible buttons on the robot's body and remote control.

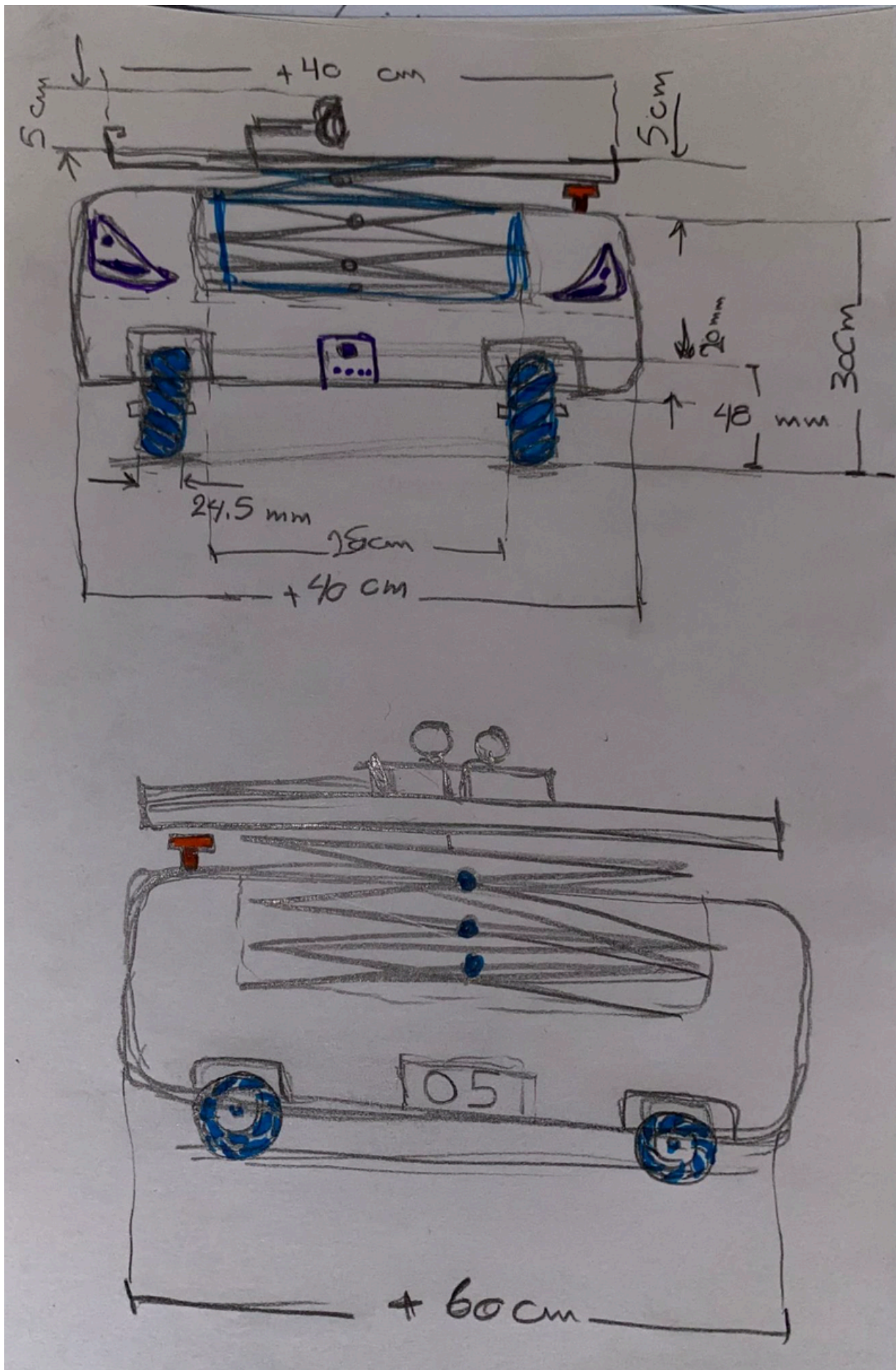
Maintenance:

- Ease of maintenance: Easy access to parts for maintenance and repairs (Modularity).
- Durability: Components must be durable to withstand industrial use

Chassis Concept design:

- Using 20x20 aluminum extrusion profile
- material: aluminum 6063 T5
- Wooden plate for the mounting of the electronics

Concept design:



Calculations:

Load Calculation:

- The load the lift needs to handle is 30 kg.
- Force (F)=Mass (m)×Gravity (g)... $F=30\text{kg}\times9.81\text{m/s}=294.3\text{N}$

For a scissor lift, the key parameters include:

- Number of scissor pairs (n)
- Length of each arm (L)
- Vertical lift height (h)
- Horizontal span of the base (b)

Assuming a single pair of scissor arms for simplicity and each arm is 400 mm long, 40cm:

- Length of each arm (L): 40 cm
- Vertical max lift height (h): 2 m
- Horizontal base span (b): +40 cm (Assume 45)

Force on the Actuator

Linear actuator:

Consider the worst-case scenario when the lift is fully collapsed:

- **Maximum Force Requirement:** When θ is very small (nearly zero), the actuator force requirement becomes maximal because the scissor arms provide the least vertical lifting force.
- **High Stress on Components:** This scenario puts the highest stress on the actuator and the scissor arms, as they need to exert maximum force to initiate the lift.

$$F_a = 294.3 \times (+40) / 2 \times 40 \times \sin\theta$$

$$F_a = 294.3 \times 0.5625$$

$$F_a = 165.54 \text{ N (depends on theta)}$$

Screw Parameters:

- Lead Screw Pitch (P): The distance the nut moves per revolution of the lead screw. For this example, let's assume it is 5 mm/rev (0.005 m/rev).
- Force (Fa): The force required by the actuator, which we previously calculated to be 165.54 N. Efficiency (η): The efficiency of the lead screw mechanism, which we'll assume to be 80% (0.8). $T = Fa \times Pitch / 2\pi \times Efficiency$

Without the linear actuator results:

$$T = \frac{F \times d_m}{2} \left(\frac{L + \mu \pi d_m}{\pi d_m - \mu L} \right)$$

- T = Torque (Nm)
- F = Load (N)
- d_m = Mean diameter of the lead screw (m)
- L = Lead (pitch) of the screw (m)
- μ = Coefficient of friction (depends on material)
- η = Efficiency, but indirectly considered through μ

Picking Mechanism :

- The suction cups must generate enough vacuum force to lift the package weight (30 kg).
- Required Force (F)=30 kg×9.81 m/s² Fa=294.3N

Suction Cup Area

- To determine the required suction cup area: Vacuum Force (F)=Pressure Difference (P)×Area (A)
- Assuming the vacuum creates a pressure difference of 0.08 MPa (80000 Pa):

$$A = \frac{F}{P}$$

$$A = \frac{294.3 \text{ N}}{80000 \text{ Pa}}$$

$$A = 0.00367875 \text{ m}^2$$

$$A \approx 36.8 \text{ cm}^2$$

If using two suction cups:

$$\text{Area per cup} = \frac{36.8 \text{ cm}^2}{2} \approx 18.4 \text{ cm}^2$$

Suction cup diameter:

$$\text{Area (A)} = \pi \times r^2$$

$$r^2 = \frac{A}{\pi}$$

$$r^2 = \frac{18.4 \text{ cm}^2}{\pi}$$

$$r^2 = \frac{18.4}{3.14159}$$

$$r \approx 2.42 \text{ cm}$$

$$\text{Diameter} \approx 2 \times 2.42 \approx 4.84 \text{ cm}$$

References:

- <https://www.shipscience.com/blog/what-are-the-different-sizes-of-amazon-boxes#:~:text=A%20Comparison%20of%20Amazon's%20Small%2C%20Medium%2C%20and%20Large%20Box%20Sizes&text=The%20medium%20box%20is%2012.5,as%20electronics%20and%20home%20goods.>
- <https://pe.usps.com/businessmail101?ViewName=Parcels#:~:text=Generally%2C%20commercial%20parcels%20must%20measure,in%20length%20and%20girth%20combined.>
- <https://www.intel.com/content/www/us/en/robotics/autonomous-mobile-robots/overview.html>

- <https://www.quasi.ai/blog-addressing-industry-challenges-with-amr-robots/#:~:text=Across%20sectors%20like%20healthcare%2C%20manufacturing,while%20keeping%20costs%20in%20check.>
- <https://www.techtargget.com/iotagenda/definition/mobile-robot-mobile-robotics>
- <https://www.earthsense.co/news/2020/7/24/levels-of-autonomy-for-field-robots>
- <https://ep-equipment.com/blog/2024/03/01/everything-about-autonomous-mobile-robots/>
- <https://adityakamath.github.io/2021-09-05-comparing-slam-methods/>
- <https://onlinelibrary.wiley.com/doi/10.1155/2020/8867937>
- <https://automaticaddison.com/navigation-and-slam-using-the-ros-2-navigation-stack/>
- https://www.researchgate.net/publication/374905560_Electromechanical_Actuator-Based_Solution_for_a_Scissor_Lift
- https://www.researchgate.net/publication/344522360_Comparison_of_different_techniques_of_2D_simultaneous_localization_and_mapping_for_a_mobile_robot_by_using_ROS
- https://answers.ros.org/question/237580/hector_slam-vs-gmapping/